

Waze User Churn Prediction

Executive Summary

This case study examines monthly user churn using behavioral activity data. The objective is to identify users who appear more likely to churn and understand the trade-offs involved in churn prediction.

Dataset & Data Quality

The raw dataset contains 14,999 records and 13 columns. The churn label is missing for 700 records, so 14,299 labeled users are used for supervised analysis. There are no fully duplicated rows in the labeled dataset. The churned class represents 17.74% of labeled observations.

Exploratory & Statistical Analysis

Users with fewer driving days show higher churn rates in the observed data, while more active users show lower churn rates. This is an observed association and does not establish causation. Android users average 66.02 drives and iPhone users average 67.93 drives. A Welch two-sample t-test gives $t = 1.6766$ and $p = 0.093651$, so the analysis does not provide sufficient evidence of a statistically significant difference at the 5% level.

Feature Engineering & Modeling

Eight behavioral features are engineered, including distance per driving day, recent-session share, total sessions per day, average recorded speed, distance per drive, favorite-navigation activity, and drives per session. The data is split into training, validation, and test sets. Outlier caps are calculated from training data only and then applied to later splits.

Logistic Regression is used as an interpretable baseline. Random Forest and XGBoost are tuned with cross validation using recall as the refit metric. XGBoost is selected as the ensemble champion because its validation recall (0.1381) exceeds Random Forest (0.1085).

Final Evaluation

Operating point	Accuracy	Precision	Recall	F1	ROC AUC
XGBoost, threshold 0.50	0.8168	0.4555	0.1716	0.2493	0.7318
XGBoost, threshold 0.223	0.7500	0.3595	0.5247	0.4266	0.7318

The 0.223 threshold was selected on validation data using an illustrative target recall of 0.50 and then frozen before testing. It increases test recall from 0.1716 to 0.5247, while precision falls from 0.4555 to 0.3595. This is an operating-point trade-off, not a universally optimal production threshold.

Key Model Signal

The current XGBoost model assigns the highest model-based importance to `activity_days` (0.249498), followed by `n_days_after_onboarding` (0.057040), `driving_days` (0.053031), `total_navigations_fav1` (0.048842), and `total_sessions_per_day` (0.045523). Feature importance describes model behavior and should not be interpreted as causal evidence.

Limitations & Next Steps

The dataset and scenario are educational. The available variables are mostly monthly aggregates, so richer drive-level and app-interaction data could provide additional signal. Before deployment, every predictor would

need to be confirmed as available at prediction time, and the operating threshold should be selected using business costs for missed churners and unnecessary outreach.